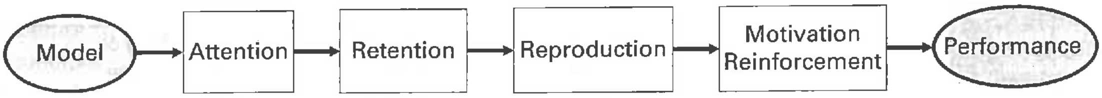
# Mechanisms of learning from a cognitive perspective

## Observational learning (modelling)

**Observational learning**, or **modelling**, occurs by watching others, noting the positive and negative consequences of their actions, and/or then imitating these actions. Observational learning is of particular benefit when trying to learn or develop a motor skill. It is often easier and more efficient to imitate an observed **model** rather than try to learn through theory, which may not be totally clear and therefore misunderstood. It is also better than using 'trial-and-error' techniques, which may lead to ongoing mistakes and/or bad habits in the implementation of the task. Furthermore, learning through observation helps us to avoid potentially threatening and harmful situations without having to directly experience them.

Observational learning is often referred to as an extension of operant conditioning as we learn through observing the consequences of others' behaviours, whether they are reinforced or punished, according to the processes outlined in operant conditioning. Because we learn from our interactions with others, Bandura also refers to his model as 'social learning theory'.

## The procedure in observational learning (as proposed by Bandura)




**Attention**: In order for something to be learnt, it must first be taken in, which requires the individual to pay attention to the model's behaviour to recognise the distinctive features of it. Furthermore, the behaviour should be clearly visible and perceived by the observer as being able to be imitated. One is more likely to attend to models that are liked, known, are similar in nature to the observer, or have a high status.

**Retention**: The observer must be able to remember the model's behaviour.

**Reproduction**: The learner must attempt to reproduce or copy what has been observed in order to demonstrate that learning has occurred. This may be dependent on physical factors within an individual and may also be influenced by environmental factors.

**Motivation reinforcement**: The learner has to be motivated to perform the behaviour in order to receive some form of reinforcement. The behaviour should have a high incentive value for the learner or act as a reward, otherwise it will be unlikely that the behaviour will be carried out. Motivation may involve external reinforcement, self-reinforcement or vicarious reinforcement.

**Vicarious classical conditioning**: Occurs when an individual, having observed another person, copies his or her reaction to a specific stimulus.

**Vicarious operant conditioning**: Refers to an individual learning by observing another individual being reinforced or punished for his or her behaviour. As a consequence the individual will either imitate this behaviour, behave in a modified manner, or refrain from engaging in the behaviour.

**Vicarious reinforcement**: Occurs when the likelihood of an observer performing a specific behaviour increases after observing another being reinforced for such a behaviour.

**Vicarious punishment**: Occurs when an observer is likely to refrain from performing a specific behaviour after observing another being punished for such a behaviour.

## Bandura's research

Bandura formulated his social learning theory after performing a series of experiments with groups of kindergarten students (equal numbers of male and female) who watched films on a television screen. Having watched the films, the children were then individually placed in a room with a one-way mirror.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The room contained several toys including a BoBo doll (an inflatable rubber clown). The researchers then observed their behaviour to see whether the children would model their own behaviour on the aggressive model to which they had attended.

In the first experiment, two groups of children observe an adult at play. For one group, the adult punches, kicks, hits and verbally abuses the BoBo doll, whereas for the second group the adult is non-aggressive in their play. A third (control) group does not observe an adult at play.

The main findings were:

- Children who observed the aggressive model made significantly more aggressive responses than those in the other two conditions.
- Boys made more aggressive responses than girls, especially if the model was male rather than female.
- The girls who observed the aggressive model also showed more physical aggressive responses if the model was male, but more verbal aggressive responses if the model was female.
- Not only were specific violent acts imitated, but the children also generalised their violence to other toys as well, for example, playing aggressively with plastic farm animals.
- The children who observed the non-aggressive model showed very little aggression, although not always significantly less than the control group.

Another experiment found that watching a film of aggressive behaviour inspired some imitation, but was less influential on children than observing the same aggressive act in person. This experiment is important, however, because it was a precedent that stimulated further studies into the effects on children of viewing violence in the media.

In a further experiment, three videos of an adult punching and abusing the BoBo doll were shown to different groups of kindergarten children. The videos differed in terms of the consequences given to the adult – rewarded, punished, or no consequence. When allowed into the playroom, those who had seen the adult rewarded or receive no consequence were more likely to imitate the aggressive behaviour, whereas those who saw the adult punished showed significantly lower levels of aggression towards the doll.

Furthermore, the children who were then offered rewards for imitating the aggressive behaviour tended to behave aggressively, including those who had seen the adult model being punished for their actions. While the boys tended to show more overall aggression than the girls, when a reward was offered the girls became nearly as aggressive as the boys. On the basis of these results, Bandura concluded that all of the kindergarten children learnt about the aggressive model's behaviour, regardless of the consequences they observed the model face. They had made cognitive representations or mental pictures of what they had learnt about behaving aggressively that could be elicited when an appropriate reward was offered. Therefore, the children acquired a learnt response which they stored until an incentive was offered.

## Factors influencing imitation of behaviour

Bandura identified several factors as being important in determining whether an individual would copy an observed behaviour. These include:

- The amount of attention that the observer paid to the model.
- The characteristics of the model, such as similarity to the observer, attractiveness and trustworthiness.
- The observer's admiration for the model, in terms of status, expertise or power.
- The observer's ability to remember what the model did (retention).
- The capabilities of the model.
- The observer's ability (or belief in his or her ability) to copy the modelled behaviour (reproduction).
- The amount of motivation that the observer had to repeat a task.
- The consequence that the model incurred as a result of the behaviour – whether the model was reinforced or punished for the behaviour.